

MUHAMMAD USMAN AL HAQ

Computer Science Student | Full-Stack Developer | Competitive Programmer
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PROFESSIONAL SUMMARY

Third-year CS student at FAST-NUCES with hands-on experience in full-stack development, mobile engineering, and HPC. ICPC Asia Topi Regionalist (2025) with strong foundations in competitive programming, system design, and production-grade software delivery.

TECHNICAL SKILLS

Languages: C++, C#, Python, Kotlin, Java, JavaScript, TypeScript, PHP, SQL, HTML/CSS
Frameworks & Libraries: Next.js, React, Node.js, Android SDK, CUDA, OpenMP, AVX2 SIMD, JavaFX, SFML, Tailwind CSS, Framer Motion
Databases: MongoDB, MySQL, PostgreSQL, SQL Server, Firebase Firestore, SQLite, Supabase
Tools & Platforms: Git, GitHub, Android Studio, VS Code, Figma, Vercel, Linux, Postman, XAMPP
Concepts: Object-Oriented Design, RESTful APIs, MVC, Parallel & Distributed Computing, ML, Agile

EDUCATION

National University of Computer & Emerging Sciences (FAST-NUCES)	Islamabad, Pakistan
Bachelor of Science in Computer Science	Aug 2023 – Present 6th Semester
International School of Pakistan (ISP)	Kuwait
HSSC Pre-Engineering – FBISE Board Score: 957/1100	2020 – 2022

PROFESSIONAL EXPERIENCE

Web Development Intern — Kuwait Kazmah Center, Hawally, Kuwait June – Aug 2025

- Redesigned company website (kkc.com.kw) end-to-end using WordPress, improving UX and visual consistency
- Designed bilingual (English & Arabic) customer feedback forms to streamline review collection

PROJECTS

Hybrid CPU+GPU Co-Execution for 2D Image Convolution / C++, CUDA, OpenMP, AVX2 SIMD, Ubuntu

- Implemented 4 variants of 15×15 separable Gaussian convolution: sequential, CUDA shared-memory, AVX2+OpenMP, and hybrid co-execution pipeline
- Profiled workload across 512×512 to 4K, identifying 40% PCIe bottleneck; proved hybrid CPU/GPU pipeline outperforms isolated GPU

CineBook Pro – Cinema Ticket Booking System / Next.js 14, MongoDB, NextAuth.js, TMDB API, Tailwind CSS

- Built full-stack cinema booking platform with interactive seat maps, role-based admin panel, and automated movie import via TMDB API
- Implemented enterprise security: bcrypt hashing, token-based password reset, JWT sessions, middleware-protected admin routes

FastPay – Enterprise Digital Wallet & P2P Ledger Ecosystem / Java 17, JavaFX, Next.js 14, Supabase, PostgreSQL, TypeScript

- Engineered full-stack fintech ecosystem: JavaFX desktop client with Next.js 14 web portal, ACID double-entry ledger, JWT auth, Postgres RLS
- Delivered virtual budget partitions, P2P/Raast fund transfers, QR payment slips, and real-time searchable grid of 26+ utility billers

List It – C2C Classified Marketplace Mobile App / Kotlin, Android Studio, Firebase, MySQL, PHP, Agora API, Google Maps

- Developed full-featured Android marketplace with real-time chat, deal negotiation, multi-image uploads, and Google Maps integration
- Implemented offline-first architecture with SQLite caching, request queuing, and sync-on-reconnect

RACE – Reading Comprehension & Quiz Generation System / Python, Scikit-learn, Streamlit, NLP, Machine Learning

- Built AI-powered system generating comprehension questions, answer verification, distractor options, and graduated hints from passages
- Trained classical ML models (LR, SVM, Naive Bayes, RF, Ensemble Voting) achieving 75% accuracy; evaluated with BLEU, ROUGE, METEOR

COMPETITIONS & ACHIEVEMENTS

- ICPC Asia Topi Regionalist 2025 – Represented FAST Islamabad at GIK Institute, clearing the qualifying round among top teams nationwide

ADDITIONAL INFORMATION

Languages: English (Professional), Urdu (Native), Arabic (Elementary) | **Certification:** Web Dev & Social Media Intern
Community: Led tree-planting initiative with Red Crescent Society Islamabad (70+ trees)